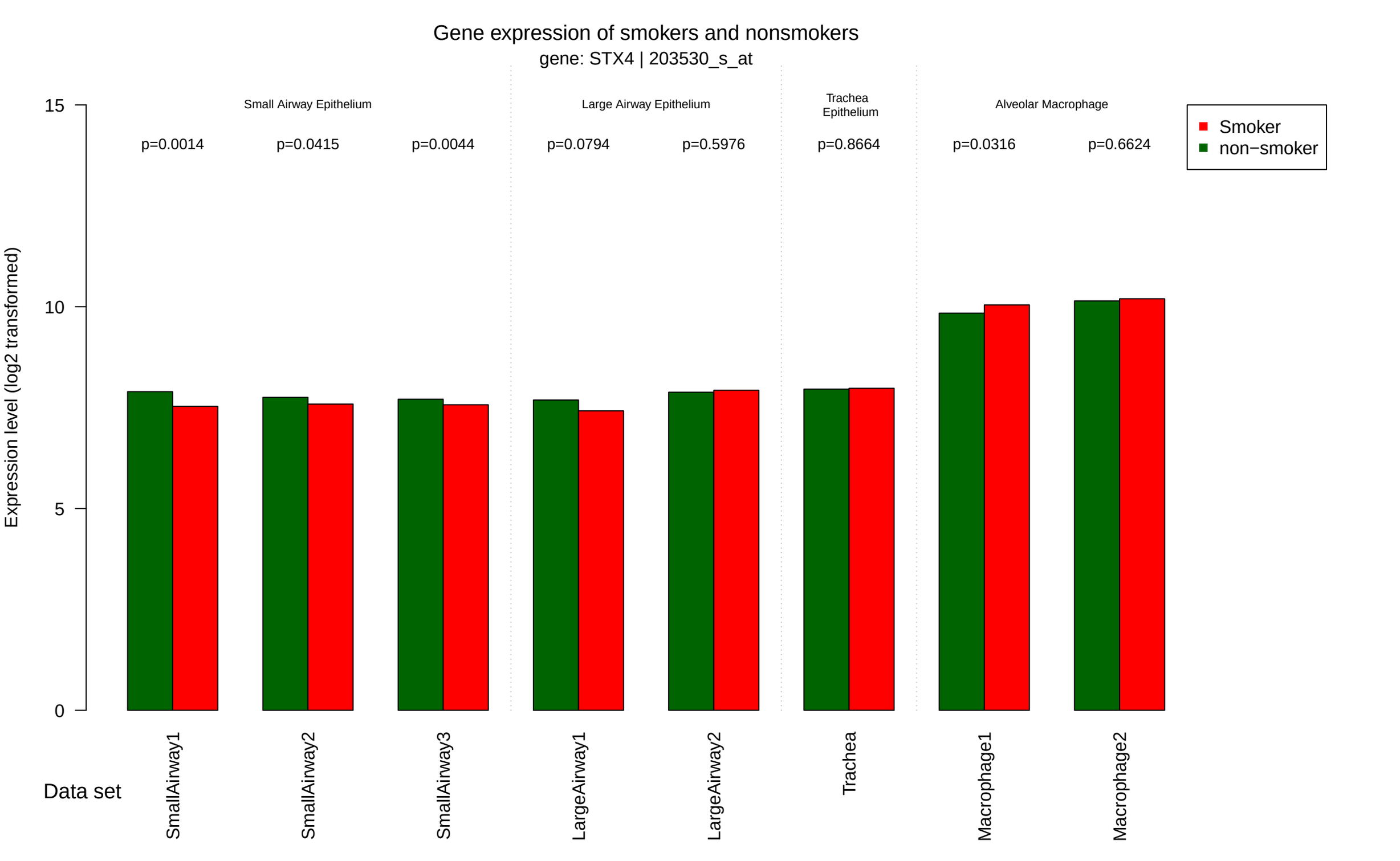
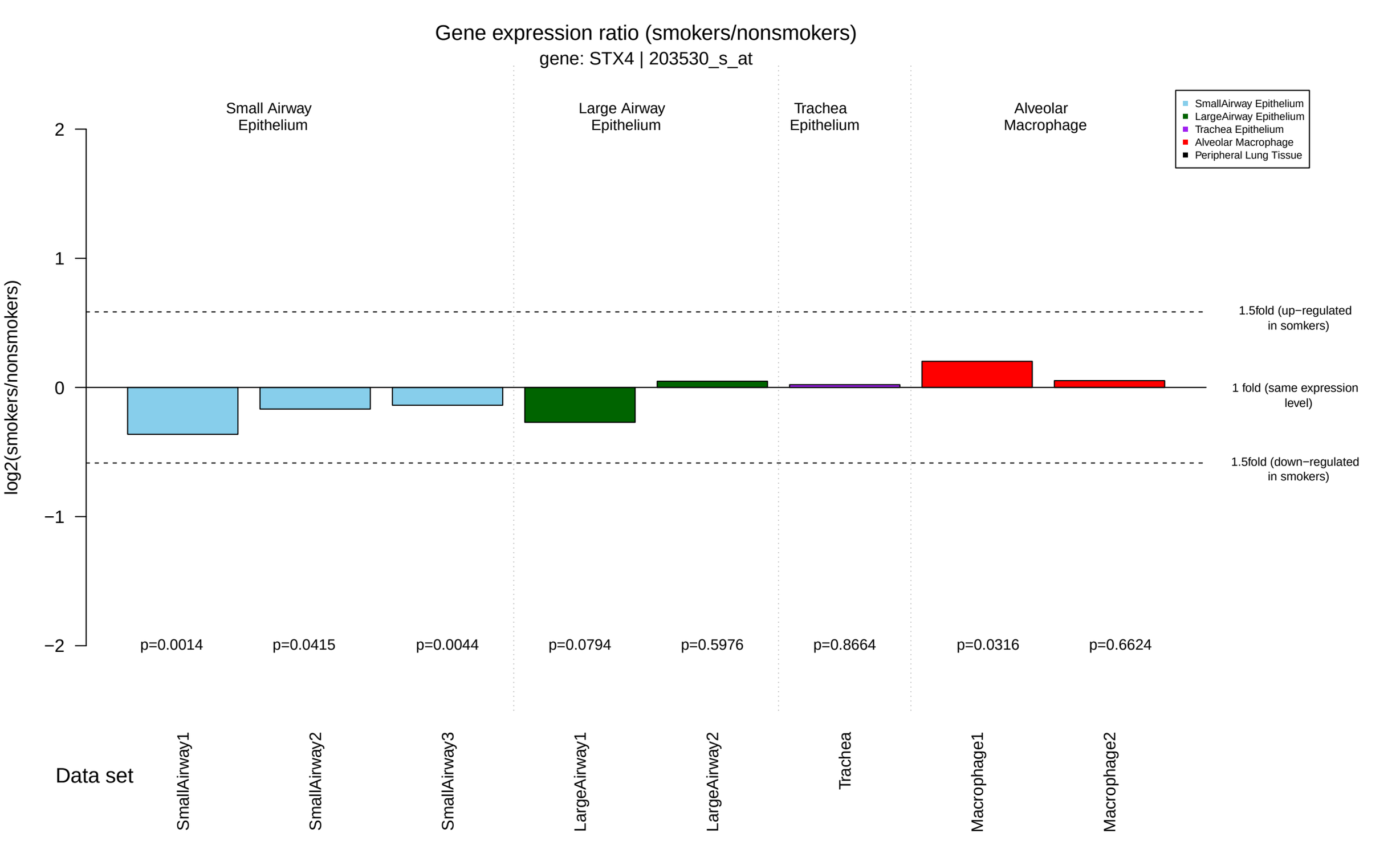
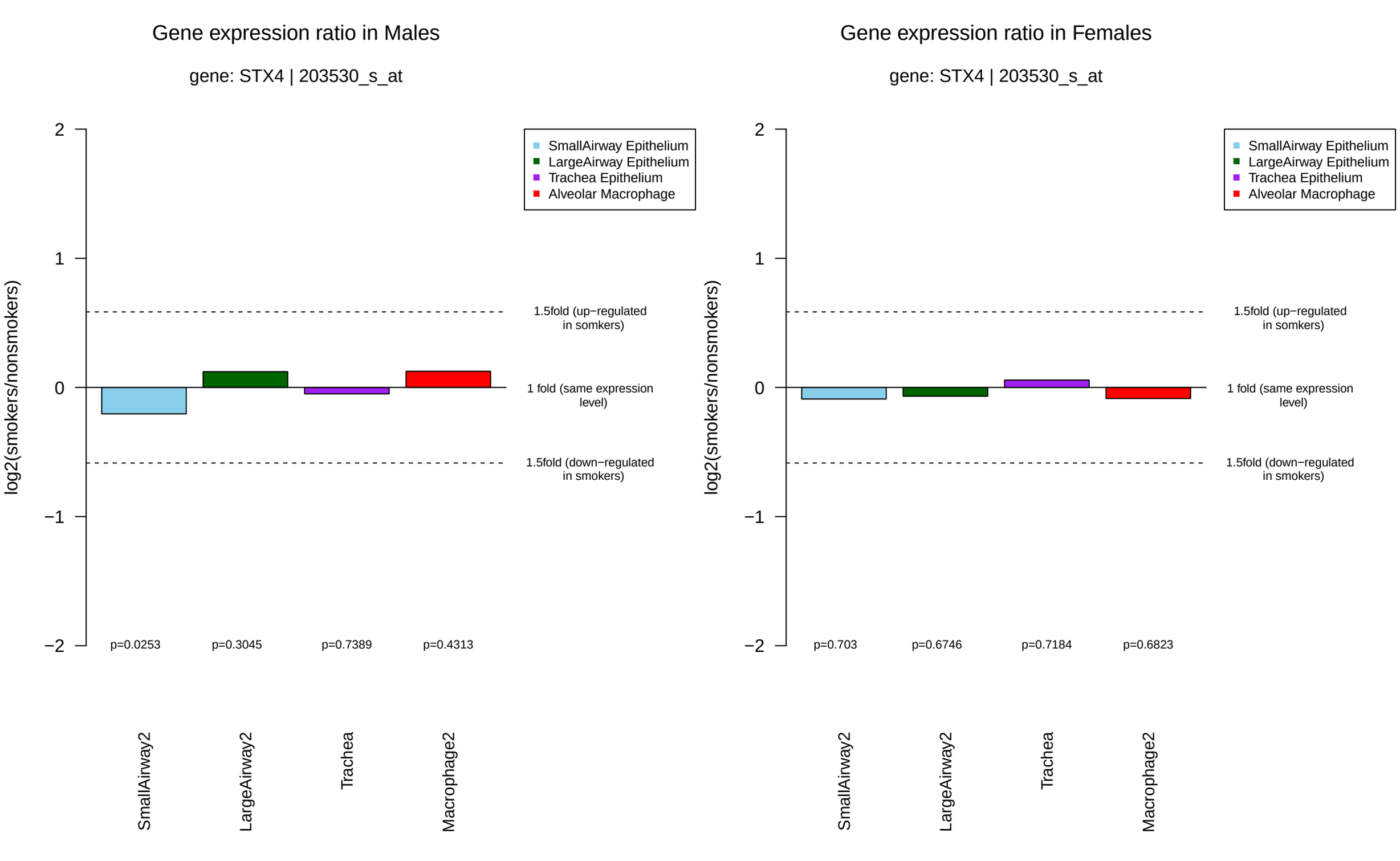
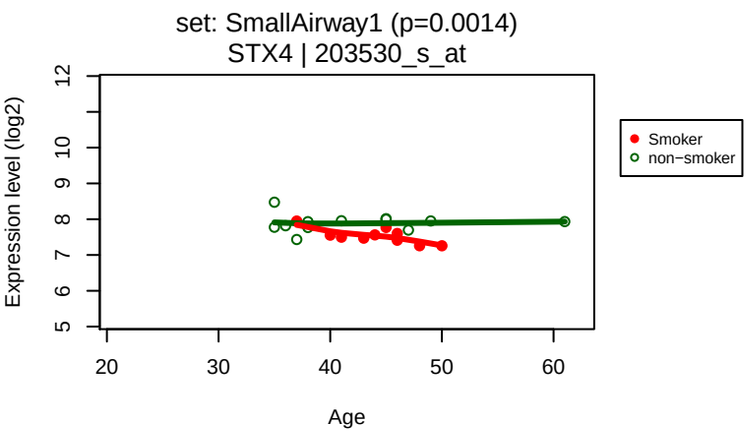
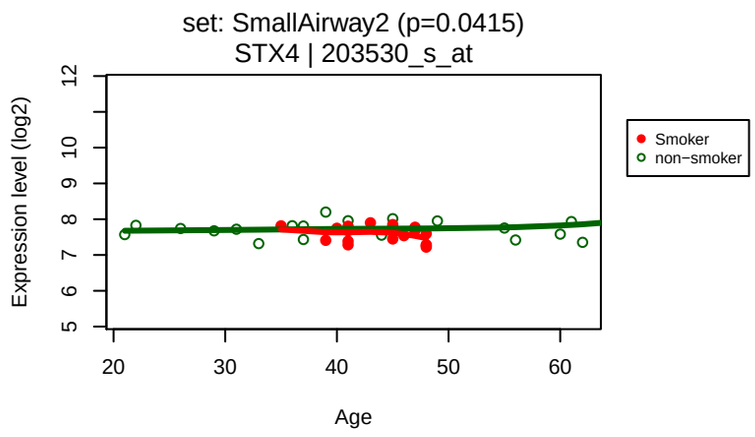
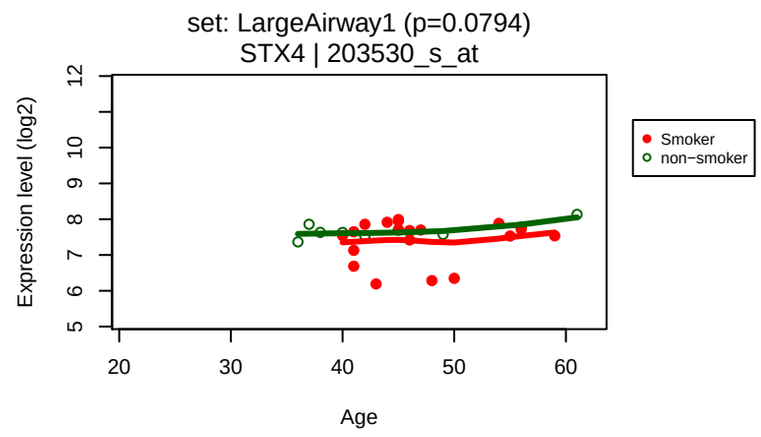

Gene expression of smokers and nonsmokers
gene: STX4 | 203530_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0014
p=0.0415
p=0.0044
p=0.0794
p=0.5976
p=0.8664
p=0.0316
p=0.6624
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

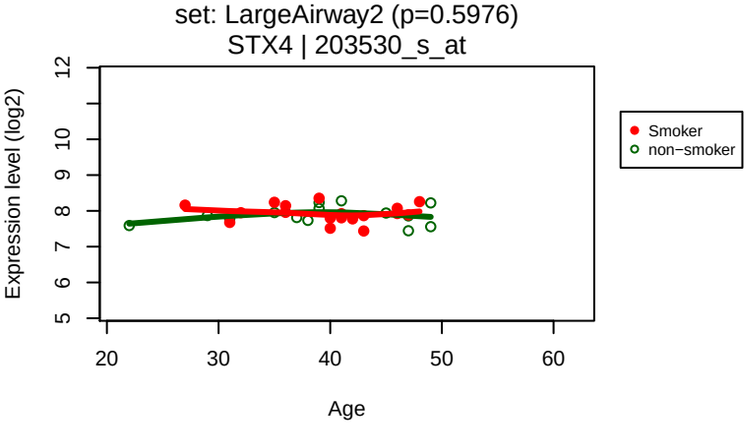

Gene expression ratio (smokers/nonsmokers)
gene: STX4 | 203530_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0014
p=0.0415
p=0.0044
p=0.0794
p=0.5976
p=0.8664
p=0.0316
p=0.6624
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

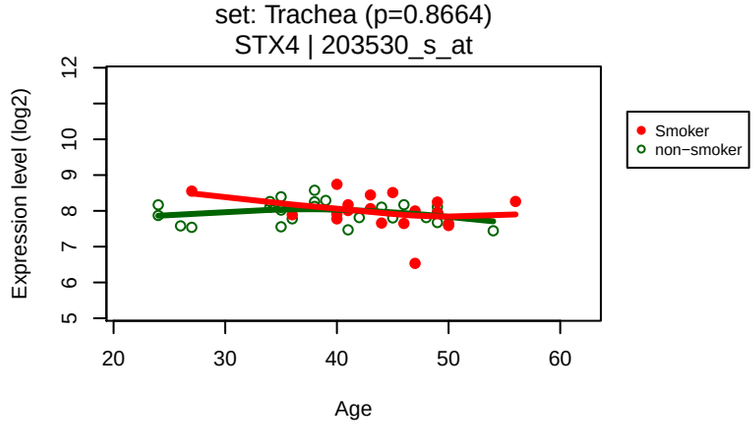

Gene expression ratio in Males
gene: STX4 | 203530_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0253
p=0.3045
p=0.7389
p=0.4313
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: STX4 | 203530_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.703
p=0.6746
p=0.7184
p=0.6823
SmallAirway2
LargeAirway2
Trachea
Macrophage2

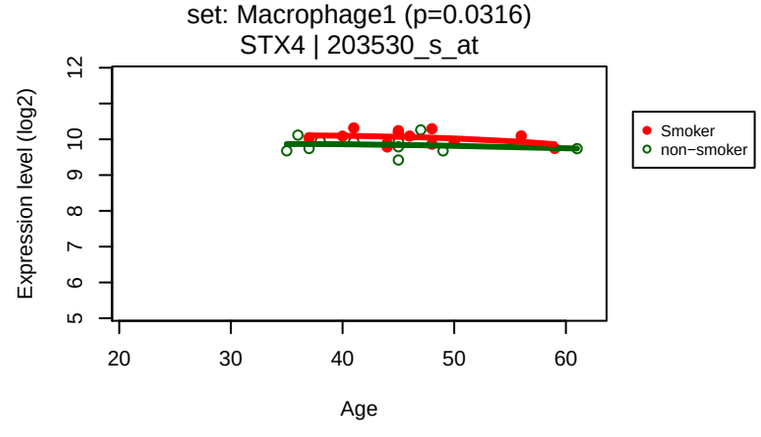

set: SmallAirway1 (p=0.0014)
STX4 | 203530_s_at
Expression level (log2)
Age
Smoker
non−smoker

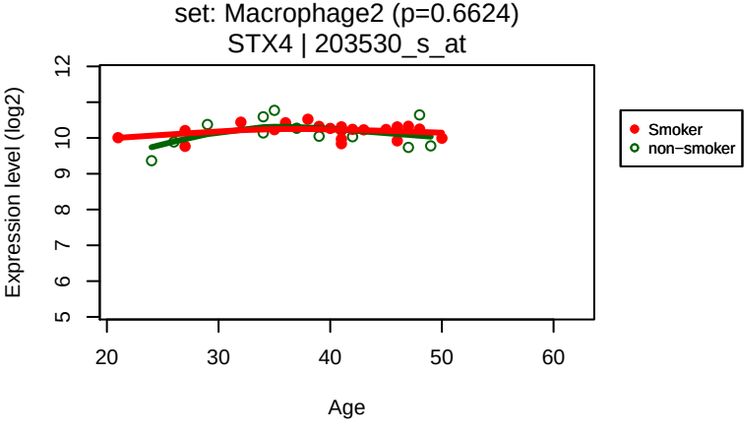

set: SmallAirway2 (p=0.0415)
STX4 | 203530_s_at
Expression level (log2)
Age
Smoker
non−smoker

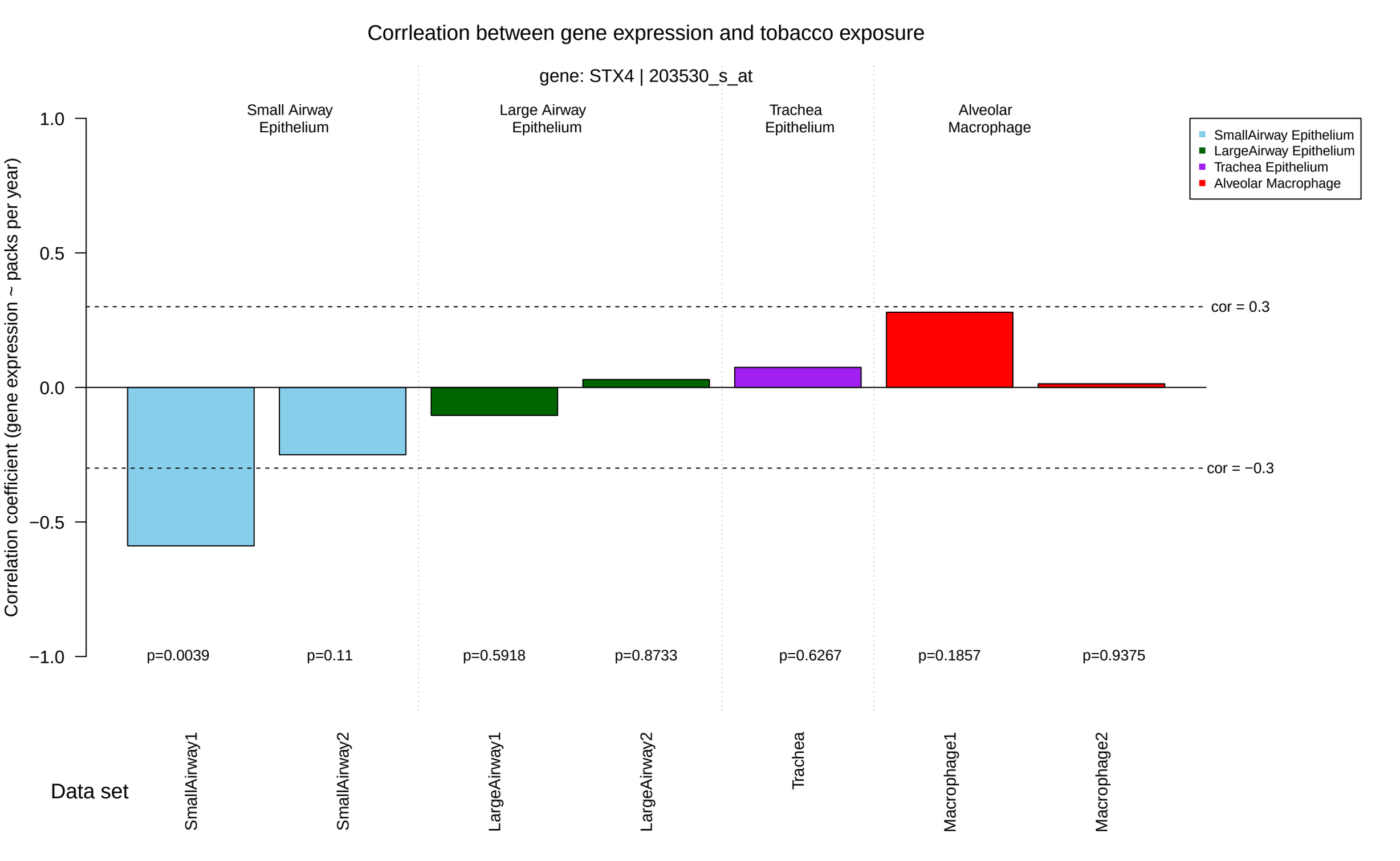

set: LargeAirway1 (p=0.0794)
STX4 | 203530_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5976)
STX4 | 203530_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8664)
STX4 | 203530_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0316)
STX4 | 203530_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6624)
STX4 | 203530_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: STX4 | 203530_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0039
p=0.11
p=0.5918
p=0.8733
p=0.6267
p=0.1857
p=0.9375
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2